

Homework #5

AMS 507, Fall 2025

Due October 9

1. Let X be a continuous random variable with pdf $f(x)$ and cdf $F(x)$. For a fixed number x_0 , define the function

$$g(x) = \begin{cases} f(x)/[1 - F(x_0)] & x \geq x_0 \\ 0 & x < x_0. \end{cases}$$

Assume $F(x_0) < 1$ and prove that $g(x)$ is a pdf.

2. Let X have the pdf $f(x) = 1/3$, $-1 < x < 2$, zero elsewhere. Find the cdf and pdf of $Y = X^2$.
3. Let X have the pdf $f(x) = (x + 2)/18$, $-2 < x < 4$, zero elsewhere. Find

- (a) $E(X)$
- (b) $E[(X + 2)^3]$
- (c) $E[6X - 2(X + 2)^3]$

4. Let X have the pdf $f(x) = 3x^2$, $0 < x < 1$, zero elsewhere.

- (a) Compute $E(X^3)$.
- (b) Find the distribution of $Y = X^3$.
- (c) Compute $E(Y)$ and compare this result with the answer obtained in part (a).

5. Let X have the pdf $f(x) = 6x(1 - x)$, $0 < x < 1$, zero elsewhere. Compute $P(\mu - 2\sigma < X < \mu + 2\sigma)$.

6. For the following cdf $F(x)$, find the pdf $f(x)$, the 25th percentile, and the 60th percentile.

- (a) $F(x) = (1 + e^{-x})^{-1}$, $-\infty < x < \infty$
- (b) $F(x) = \frac{1}{2} + \frac{1}{\pi} \tan^{-1}(x)$, $-\infty < x < \infty$

7. Let X have a r.v. with mean $\mu = E(X)$. Then for any real number b , show that

$$E[(X - b)^2] = E[(X - \mu)^2] + (\mu - b)^2 = \text{Var}(X) + (\mu - b)^2$$

and hence $E[(X - b)^2]$ is minimized when $b = \mu$, the mean of X .

8. Let X be a continuous r.v. with pdf $f(x)$. If m is the unique median of the distribution of X , and a is any real number, show that

$$E(|X - a|) \geq E(|X - m|),$$

provided the expectations exist.

9. Suppose the survival time (in week) of a certain strain of mice has a distribution with cdf

$$F(t) = 1 - \exp \left\{ -\eta \delta_1 (t/104)^{\delta_2} \right\},$$

where η , δ_1 and δ_2 are constants. Show how to generate a random number from this distribution starting with a Uniform random number.

10. In relation to continuous nonnegative random variables T that might represent lifetimes or times to failure, with pdf $f(t)$ and cdf $F(t)$, we define the failure rate function as

$$\lambda(t) = \lim_{\Delta t \rightarrow 0} \frac{1}{\Delta t} P(t < T \leq t + \Delta t | T > t).$$

- (a) For the Weibull pdf given by $f(t) = \beta \lambda (\lambda t)^{\beta-1} \exp[-(\lambda t)^\beta]$, $t > 0$, with parameters $\lambda > 0$ and $\beta > 0$, obtain the failure rate function $\lambda(t)$, $t > 0$.
- (b) Show that $F(t) = 1 - \exp \left[- \int_0^t \lambda(u) du \right]$ and $f(t) = \lambda(t) \exp \left[- \int_0^t \lambda(u) du \right]$, $t > 0$ in general.

11. A standard Cauchy random variable has density function

$$f(x) = \frac{1}{\pi(1+x^2)}, \quad -\infty < x < \infty.$$

If X is a standard Cauchy random variable, show that $Y = 1/X$ is also a standard Cauchy random variable if $f_Y(0) = 1/\pi$.